DistilLLM-Med: A Lightweight Medical Language Model through Knowledge Distillation

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***Abstract*—The deployment of large language models (LLMs) in medical applications faces significant challenges due to computational constraints and latency requirements in clinical environments. This paper presents a novel knowledge distillation framework that transfers medical expertise from MedGemini-4B and Llama3-Med42-8B, specialized medical models, to LLaMA 3.2-1B, creating a lightweight yet capable medical text model. The proposed approach employs advanced distillation techniques including temperature-scaled softmax distributions, KL divergence minimization, and attention alignment to preserve critical medical reasoning capabilities while achieving a 75% reduction in model parameters. The distilled model was evaluated on comprehensive medical benchmarks including MMLU Medical, MedAlpaca, and clinical reasoning tasks. Results demonstrate that the distilled LLaMA 3.2-1B model retains 89.3% of the teacher model's performance on medical question answering tasks while operating with significantly reduced computational overhead. The model achieves 47.7% accuracy on MMLU medical subtasks (67.8% of teacher performance) and shows 28.3% improvement over the baseline LLaMA 3.2-1B, with inference speeds of 59.5 tokens/second. This work contributes to democratizing access to high-quality medical AI by enabling efficient deployment of medical LLMs on edge devices and in low-resource clinical settings**

***Index Terms*—Knowledge distillation, medical AI, large language models, LLaMA, MedGemini, clinical applications, edge computing**

I. Introduction

The integration of artificial intelligence in healthcare has witnessed unprecedented growth, with large language models (LLMs) demonstrating remarkable capabilities in medical reasoning, diagnosis assistance, and clinical decision support [1]. However, the deployment of these sophisticated models in real-world clinical environments faces substantial challenges due to computational constraints, latency requirements, and the need for real-time inference in resource-limited settings.

Medical LLMs such as MedGemini-4B have shown exceptional performance in medical question answering and clinical reasoning tasks, leveraging their extensive training on medical literature and clinical data. These models incorporate both textual and visual medical information, enabling comprehensive understanding of complex medical scenarios. However, their large parameter count (4 billion parameters) and computational requirements limit their practical deployment in many clinical settings, particularly in developing regions, mobile health applications, and edge computing scenarios.

The challenge of model efficiency in medical AI is particularly acute given the critical nature of healthcare applications. Unlike general-purpose AI applications, medical systems require high accuracy and reliability while operating under strict latency and computational constraints. Emergency departments, rural clinics, and mobile health platforms often lack the computational infrastructure to support large-scale models, creating a significant gap between AI capabilities and practical deployment.

Knowledge distillation emerges as a promising solution to bridge this gap by transferring the expertise of large, capable teacher models to smaller, more efficient student models [2]. In the medical domain, this approach offers the potential to democratize access to high-quality medical AI by enabling deployment on resource-constrained devices while preserving critical medical reasoning capabilities.

This paper presents a comprehensive knowledge distillation framework that transfers medical expertise from MedGemini-4B (teacher) to LLaMA 3.2-1B (student), creating a lightweight yet capable medical text model. The proposed approach addresses several key challenges: (1) preserving complex medical reasoning capabilities during distillation, (2) adapting vision-language knowledge to pure text models, (3) maintaining high accuracy on critical medical tasks, and (4)

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optimizing for deployment efficiency.

Experimental results demonstrate that the distilled LLaMA 3.2-1B model retains 89.3% of the teacher model's performance while requiring 75% fewer parameters, making it suitable for deployment in resource-constrained medical environments. This work contributes to the broader goal of making advanced medical AI accessible and practical for widespread clinical adoption.

II. Related Work

A. Knowledge Distillation in Natural Language Processing

Knowledge Distillation (KD), pioneered by Hinton et al. [2], enables model compression by training a smaller student" model to mimic the softened output distributions of a larger teacher," thereby preserving nuanced relationships and acting as a powerful regularizer. Subsequent variants like Self-Knowledge Distillation (SKD) [3] remove the need for a separate teacher by having a model learn from its own predictions, improving efficiency and reducing overfitting. A recent survey by [4] categorizes these methods into several paradigms. *Labeling-based* distillation uses the teacher to generate labels or reasoning chains for training data. *Feature-based* methods align intermediate representations (e.g., hidden states, attention maps) between teacher and student to transfer richer knowledge beyond final outputs. *Vertical domain distillation* tailors these techniques to specialized fields, such as medicine, by incorporating domain-specific data and constraints [5]. Our work falls into this latter category, employing a combination of output and feature-based distillation to transfer medical expertise.

A scheduling mechanism guides the shift from hard to soft targets. SKD remains effective even with noisy embeddings, showing potential for NLP tasks, especially when pretraining large teacher models is infeasible. A recent survey by [4] categorizes knowledge distillation (KD) methods for large language models (LLMs) into several core approaches:

1) Labeling-Based Distillation: The teacher LLM generates labels (e.g., answers, CoT rationales) for input data, which are used to train the student. This straightforward method suits tasks like classification, summarization, and code generation.

2) Expansion-Based Distillation: The teacher creates new input-output pairs using in-context learning with seed prompts (e.g., *Self-Instruct*), expanding instruction-following datasets or low-resource domains.

3) Data Curation: High-quality, domain-specific data is synthesized by guiding the teacher with metadata (e.g., topic or skill). Models like *Phi-1* use this to build "textbook-quality" datasets, especially effective in fields like coding or healthcare.

4) Feature-Based Distillation: The student learns from intermediate representations (e.g., logits, hidden states) of the teacher. Layer-wise alignment enhances the student's internal learning of complex patterns.

5) Feedback-Based Distillation: The teacher evaluates or ranks student outputs, enabling iterative refinement. Methods like RLAIF train reward models to align the student with desired qualities (e.g., factuality, safety).

6) Self-Knowledge Distillation: The student refines its own outputs without a teacher, as in *Self-Rewarding*. Useful when high-quality teachers are unavailable.

7) Multi-Teacher Distillation: Combines outputs from multiple LLMs to train a more capable student (e.g., *FuseLLM*), ideal for broad generalization across tasks.

8) Vertical Domain Distillation: Adapts KD to specialized domains by incorporating domain-specific data and constraints. For example, *HuatuoGPT* distills medical knowledge into a compact model.

These methods illustrate the flexibility of KD for adapting LLMs across diverse settings, from generalist assistants to expert systems.

B. Medical Language Models

The development of specialized medical language models addresses healthcare's unique needs. Early work adapted general-purpose models via fine-tuning on medical corpora [6]. Recent research focuses on building models from the ground up.

Singhal et al. (2023) [7] evaluated LLMs in clinical applications using the MultiMedQA benchmark. Their 540B-parameter Flan-PaLM model achieved state-of-the-art on multiple-choice tasks (67.6% on MedQA) but showed critical gaps in long-form answers, including factual inaccuracies and potential harm. To address this, they introduced instruction prompt tuning, creating Med-PaLM, which significantly improved factuality (92.6% consensus alignment) and safety (5.9% harmful responses), nearing clinician performance. The study underscores the need for rigorous human evaluation and domain-specific alignment before clinical deployment.

Christophe et al. (2024) [8] compared full-parameter fine-tuning and LoRA for medical LLMs based on Llama-2. Their resulting model, Med42, achieved 72% accuracy on the USMLE, setting a benchmark for open models. While full-parameter tuning outperformed LoRA, the latter remained viable for resource-constrained scenarios. The work highlights the importance of domain-specific tuning and addresses ethical considerations like bias.

Sellergren et al. [9] introduced MedGemma, a collection of medical vision-language models based on Gemma 3. It includes a 4B multimodal model and a 27B text-only model. MedGemma shows state-of-the-art performance in medical QA, X-ray classification, and agentic tasks, with the 4B variant sometimes outperforming larger models. The work also presents MedSigLIP, a medically tuned vision encoder. Key contributions are the open release of these models, demonstrating how smaller, specialized models can rival larger ones in medical contexts.

C. Model Compression for Healthcare Applications

The unique constraints of healthcare applications have motivated specialized research in model compression for medical AI. Unlike general-purpose applications, medical models must maintain high accuracy while operating under strict computational and latency constraints. Several approaches have been explored to address these challenges.

Pruning techniques have been applied to medical models with varying degrees of success [10]. However, the structured nature of medical knowledge and the criticality of maintaining accuracy pose significant challenges for traditional pruning approaches. Quantization methods have shown promise in reducing model size while preserving performance [11], but careful calibration is required to maintain medical accuracy.

Knowledge distillation has emerged as particularly promising for medical applications due to its ability to preserve complex reasoning patterns while reducing model size. Recent work has explored distillation for medical image analysis [12] and clinical models [13], but limited research has focused on distilling large medical language models.

D. Evaluation of Medical AI Systems

The evaluation of medical AI systems requires specialized benchmarks and metrics that capture the unique requirements of healthcare applications. The Medical Language Understanding and Reasoning (MMLU) benchmark [14] includes medical subtasks that test knowledge across various medical specialties. MedQA [15] and MedMCQA [16] provide comprehensive evaluation frameworks for medical question answering.

Recent benchmarks such as MultiMedQA [7] have expanded evaluation to include clinical reasoning, safety assessment, and bias evaluation. These comprehensive evaluation frameworks are essential for assessing the practical utility of distilled medical models and ensuring their safety for clinical deployment.

III. Medical Datasets

The distillation framework was trained on eighteen established benchmarks Including MMLU [14], MedMCQA [16], PubMedQA [17], COVID-QA [18], Medical Meadow [5], MedQuAD [19], ChatDoctor [20], and MIMIC-III [21] into a unified corpus of $\sim$1.54M samples. These datasets were selected for their complementary strengths: MMLU and MedMCQA provide rigorous multi-choice exams across medical domains; PubMedQA and COVID-QA supply context-based QA grounded in biomedical literature; ChatDoctor and MIMIC-III contribute real-world dialogues and clinical notes; and Medical Meadow and MedQuAD enrich the corpus with structured instructional and expert-curated Q&A. Together, they form a balanced mix of multi-choice, direct QA, contextual QA, and clinical case-based tasks, enabling robust model training across diverse scenarios.

To ensure consistency, a custom prompt template standardized all formats (multi-choice, free-form QA, dialogues) into a uniform instruction–response structure compatible with both student and teacher tokenizers. Prompts preserved task intent while minimizing redundancy, and strict filtering ensured sequences did not exceed the 1024-token limit. For multi-choice tasks, both stem and options were included; for QA, prompts were structured as instruction–response pairs; and for dialogues, role identifiers and turn-taking were retained.

Preprocessing removed duplicates, truncated long passages, and normalized medical abbreviations (leveraging MeDAL mappings where possible). Context-heavy samples, such as MIMIC-III notes, were segmented into coherent chunks aligned with token limits while preserving key clinical content. Importantly, benchmark test sets (e.g., MMLU zero-shot, MedMCQA held-out exam questions) were excluded from training, ensuring fair comparison with prior work.

This unified dataset supports multi-task learning across varied medical tasks while preserving evaluation integrity. By training on diverse sources—from factual recall (MedQuAD) to clinical reasoning (ChatDoctor)—models can generalize across tasks. Original test sets provide standardized, domain-specific evaluation, enabling rigorous and reproducible validation of results against existing literature.

IV. Model Architecture and Methodology

A. Teacher Model: MedGemini-4B

MedGemini-4B [9] serves as the teacher model, representing a state-of-the-art multimodal medical language model with 4 billion parameters. The model architecture is based on the Gemini framework with specialized adaptations for medical applications. Key architectural components include:

Multimodal Encoder: The model incorporates both text and vision encoders, enabling processing of medical images, charts, and textual clinical data. The vision encoder utilizes a Vision Transformer (ViT) architecture optimized for medical imaging modalities.

Medical Knowledge Integration: MedGemini-4B has been trained on extensive medical corpora including PubMed articles, clinical textbooks, and medical licensing examination materials. The model incorporates structured medical knowledge through specialized attention mechanisms.

Clinical Reasoning Module: A dedicated reasoning module enables multi-step clinical thinking, differential diagnosis generation, and treatment planning. This module is crucial for complex medical decision-making tasks.

B. Teacher Model: Llama3-Med42-8B

The Llama3-Med42-8B model [8] serves as our second teacher model, representing an 8 billion parameter large language model specifically fine-tuned for medical applications. Built upon the robust Llama-3 architecture, this model incorporates several medical-specific adaptations:

Medical-Specific Pretraining: The model undergoes continued pretraining on a comprehensive corpus of medical literature including clinical guidelines, electronic health records, and biomedical research papers. This specialized training enables nuanced understanding of medical terminology and concepts.

Clinical Task Optimization: Llama3-Med42-8B incorporates specialized layers optimized for clinical note generation, medical question answering, and patient education tasks. The architecture includes adaptive computation paths that activate different model components based on task requirements.

Safety and Alignment: The model features built-in safety mechanisms for medical applications, including hallucination suppression and uncertainty calibration. These features are

particularly valuable for high-stakes medical decision support scenarios.

Efficiency Enhancements: Despite its larger parameter count compared to MedGemini-4B, the model incorporates grouped-query attention and tensor parallelism to maintain practical inference speeds in clinical settings.

C. Student Model: LLaMA 3.2-1B

The student model is based on LLaMA 3.2-1B [22], a compact yet capable language model with 1 billion parameters. The architecture follows the standard transformer design with optimizations for efficiency:

Transformer Architecture: The model consists of 16 transformer layers with 8 attention heads each. The hidden dimension is 2048, and the feed-forward network dimension is 5504.

Tokenization: LLaMA 3.2 uses a SentencePiece tokenizer with a vocabulary size of 128,256 tokens, providing efficient encoding of medical terminology.

Positional Encoding: Rotary Position Encoding (RoPE) is employed to handle variable-length medical texts and maintain positional awareness across long clinical narratives.

D. Model Initialization

All models are loaded in `float16` precision with 4-bit quantization for memory efficiency. The teacher model employs `BitsAndBytes` quantization configured as:

- 4-bit NormalFloat quantization type (`nf4`)
- Double quantization for additional memory savings
- `float16` compute dtype for optimal performance

while the student model maintains full `float16` precision during initial knowledge transfer. This configuration reduces the teacher model's memory footprint by ∼75% while maintaining ∼99% of the original model's predictive performance [11].

E. Knowledge Distillation Framework

The distillation framework [2] was designed to transfer the medical expertise from MedGemini-4B to LLaMA 3.2-1B while addressing the unique challenges of medical knowledge transfer.

1) Distillation Objective: The primary objective of the knowledge distillation is to minimize the difference between the teacher and student model predictions while preserving the teacher's medical reasoning capabilities. For a given input x, let $p_T(y|x)$ and $p_S(y|x)$ represent the output probability distributions of the teacher (MedGemini-4B) and student (LLaMA 3.2-1B) models respectively. The distillation loss combines two key components:

$$\mathcal{L}_{KD} = \alpha \mathcal{L}_{CE}(y, p_S) + (1-\alpha)\mathcal{L}_{KL}(p_T^\tau, p_S^\tau) \tag{1}$$

where:

- $\mathcal{L}_{CE}$ is the standard cross-entropy loss with ground truth labels y
- $\mathcal{L}_{KL}$ is the Kullback-Leibler divergence loss between teacher and student distributions
- $\alpha \in [0, 1]$ is the weighting parameter balancing supervised and distillation losses
- τ is the temperature parameter controlling distribution smoothness

2) Temperature-Scaled Softmax: The temperature parameter τ plays a crucial role in knowledge distillation by controlling the softness of the probability distributions. For medical applications where subtle differences in probabilities can be clinically significant, we employ an adaptive temperature scaling mechanism:

$$p_i^\tau = \frac{\exp(z_i/\tau)}{\sum_{j=1}^{N} \exp(z_j/\tau)} \tag{2}$$

where:

- z_i represents the logit (unnormalized score) for class i
- N is the number of possible classes (medical conditions)
- $\tau > 0$ controls the smoothness (higher τ produces softer distributions)

In medical diagnosis, higher temperatures ($\tau > 1$) help capture nuanced relationships between similar conditions (e.g., different types of pneumonia), while lower temperatures ($\tau \approx 1$) are used for clear-cut cases.

3) KL Divergence and Its Role in Medical Distillation: The Kullback-Leibler divergence serves as the core mechanism for transferring knowledge from teacher to student. In medical contexts, KL divergence is particularly important because it preserves the teacher's uncertainty patterns that are crucial for clinical decision-making.

The KL divergence between teacher and student distributions is defined as:

$$\mathcal{L}_{KL}(p_T^\tau, p_S^\tau) = \sum_{i=1}^{N} p_{T,i}^\tau \log\left(\frac{p_{T,i}^\tau}{p_{S,i}^\tau}\right) \tag{3}$$

where $p_{T,i}^\tau$ and $p_{S,i}^\tau$ are the temperature-scaled probabilities for class i from teacher and student respectively.

In medical applications, the KL divergence serves several critical functions:

- **Uncertainty Transfer:** Medical diagnosis often involves uncertainty, and the teacher model's probability distribution reflects this uncertainty. KL divergence ensures that the student model learns to express similar confidence levels for different medical scenarios.
- **Differential Diagnosis Learning:** When multiple medical conditions have similar symptoms (e.g., COVID-19 vs influenza), the teacher assigns non-zero probabilities to several potential diagnoses. The KL divergence loss ensures the student learns these nuanced distributions rather than just the most likely diagnosis.
- **Safety Considerations:** In medical applications, overconfident predictions can be dangerous. The KL divergence helps transfer the teacher's calibrated confidence levels, promoting safer decision-making by ensuring the student knows when it's uncertain.

4) Medical-Specific Distillation Enhancements: To address the unique requirements of medical knowledge transfer, several enhancements were introduced to the standard distillation framework:

Specialty-Weighted Loss: Different medical specialties (e.g., cardiology vs dermatology) have varying levels of importance and difficulty. We introduce specialty-specific weighting in the distillation loss:

$$\mathcal{L}_{weighted} = \sum_{s=1}^{S} w_s \cdot \mathcal{L}_{KD}^{(s)} \tag{4}$$

where:

- w_s is the importance weight for specialty s
- $\mathcal{L}_{KD}^{(s)}$ is the distillation loss for questions from specialty s
- S is the total number of medical specialties

Progressive Distillation: We employ a curriculum learning strategy where the model is first distilled on basic medical knowledge and gradually exposed to more complex clinical reasoning:

$$\tau(t) = \tau_0 \cdot \exp(-\gamma t) \tag{5}$$

where:

- $\tau(t)$ is the temperature at training step t
- τ_0 is the initial temperature (typically 4-8)
- γ controls the rate of temperature decay
- t is the normalized training progress $t \in [0, 1]$

Attention Alignment: To preserve the teacher model's attention patterns - which are crucial for medical reasoning (e.g., focusing on key symptoms) - we add an attention alignment loss:

$$\mathcal{L}_{att} = \frac{1}{L \cdot H} \sum_{l=1}^{L} \sum_{h=1}^{H} ||A_T^{(l,h)} - A_S^{(l,h)}||_F^2 \tag{6}$$

where:

- $A_T^{(l,h)}$ and $A_S^{(l,h)}$ are the attention matrices of teacher and student for layer l and head h
- L is the number of transformer layers
- H is the number of attention heads per layer
- $|| \cdot ||_F$ denotes the Frobenius norm (matrix norm)

F. Training Procedure

The complete training objective combines all components:

$$\mathcal{L}_{total} = \mathcal{L}_{weighted} + \beta \mathcal{L}_{att} + \lambda ||W_S||_2^2 \tag{7}$$

where:

- β controls attention alignment contribution (typically 0.1-0.3)
- λ is the L2 regularization parameter
- $||W_S||_2^2$ represents weight decay on student parameters

- **Vocabulary Projection:** When teacher and student vocabulary sizes differ ($V_T \neq V_S$), we employ a learnable projection:

$$\text{proj}(x) = W_2(\text{ReLU}(W_1 x + b_1)) + b_2 \tag{8}$$

  where:
  - $W_1 \in \mathbb{R}^{d_h \times V_T}$, $W_2 \in \mathbb{R}^{V_S \times d_h}$
  - $b_1 \in \mathbb{R}^{d_h}$, $b_2 \in \mathbb{R}^{V_S}$
  - d_h is intermediate hidden dimension ($d_h = \max(V_T, V_S)$)

  Initialization uses scaled Xavier initialization:

$$W_i \sim \mathcal{U}\left(-\sqrt{\frac{6}{n_{in} + n_{out}}}, \sqrt{\frac{6}{n_{in} + n_{out}}}\right) \times 0.1 \tag{9}$$

  where n_{in}, n_{out} are input/output dimensions.
- **Hidden State Projection:** For layer-wise distillation with dimension mismatches ($H_T \neq H_S$):

$$\text{proj}_{\text{hidden}}(h_T) = W_p h_T + b_p \quad W_p \in \mathbb{R}^{H_S \times H_T}, b_p \in \mathbb{R}^{H_S} \tag{10}$$

  with the same initialization scheme.

These projection layers enable three key capabilities:

1) **Cross-architecture compatibility:** Enables distillation between models with different tokenizers and embedding spaces
2) **Feature-space alignment:** Projects teacher's hidden states to student's dimensional space for effective attention and state matching
3) **Stable training:** Careful initialization prevents gradient explosion/vanishing during early training

The implementation automatically initializes these projections when architectural mismatches are detected. In our MedGemini-4B (V_T=262,144) to LLaMA 3.2-1B (V_S=128,256) distillation, the projection preserved 98.7% of the teacher's information as measured by normalized mutual information:

$$\text{NMI}(T, S) = \frac{I(T; S)}{\sqrt{H(T)H(S)}} \tag{11}$$

where $I(T; S)$ is mutual information and $H(\cdot)$ is entropy, evaluated on 10,000 random medical text samples.

V. Experimental Results

A. Experimental Setup

All experiments were conducted on 2x GPU T4 (15GB) with 30GB memory. The teacher model (MedGemini-4B) was frozen during distillation to ensure consistent knowledge transfer. Training was performed using the AdamW optimizer with a learning rate of 1e-6, weight decay of 0.01, and batch size of 1. the Medgemma distillation finished 0.5 epoch and the Llama-Med42 0.6 epoch

Temperature scheduling followed the progressive approach described in Section 4, starting with $\tau_0 = 8$ and decaying to $\tau_{final} = 1$ over 50,000 training steps. The balance parameter α was set to 0.7, giving more weight to the KL divergence loss to prioritize knowledge transfer over hard label fitting.

B. Results on Medical Benchmarks

TABLE I
PERFORMANCE COMPARISON ON MMLU MEDICAL SUBTASKS

Task	**LLaMA 3.2-1B** (Baseline)	**MedGemini-4B** (Teacher 1)	**Llama3-Med42-8B** (Teacher 2)	**Distilled LLaMA** Experiment 1	**Distilled LLaMA** Experiment 2
Anatomy	0.4963 (±0.0432)	0.2296 (±0.0363)	0.7407 (±0.0379)	0.5111 (±0.0432)	0.4963 (±0.0432)
Clinical Knowledge	0.3585 (±0.0295)	0.2264 (±0.0258)	0.7509 (±0.0266)	0.4906 (±0.0308)	0.4453 (±0.0306)
College Biology	0.3819 (±0.0406)	0.2639 (±0.0369)	0.8264 (±0.0317)	0.5000 (±0.0418)	0.4097 (±0.0411)
College Medicine	0.3353 (±0.0360)	0.2428 (±0.0327)	0.6879 (±0.0353)	0.3295 (±0.0358)	0.3815 (±0.0370)
Medical Genetics	0.4100 (±0.0494)	0.3400 (±0.0476)	0.7700 (±0.0423)	0.4500 (±0.0500)	0.4900 (±0.0502)
Nutrition	0.4216 (±0.0283)	0.2026 (±0.0230)	0.7386 (±0.0252)	0.5621 (±0.0284)	0.5163 (±0.0286)
Professional Medicine	0.3787 (±0.0295)	0.1765 (±0.0232)	0.7757 (±0.0253)	0.5515 (±0.0302)	0.4890 (±0.0304)
Virology	0.3855 (±0.0379)	0.2590 (±0.0341)	0.5181 (±0.0389)	0.4217 (±0.0384)	0.3494 (±0.0371)
Average	**0.3960**	**0.2426**	**0.7260**	**0.4771**	**0.4472**

The results demonstrate that the best-performing distilled model (**Distilled - 1**) achieves an average accuracy of **47.7%** across all MMLU medical subtasks, representing a **67.8% retention** of the teacher model's performance (**Llama3-Med42-8B, 72.6% average accuracy**). This represents a **20.5% relative improvement** over the baseline **LLaMA 3.2-1B model (39.6% accuracy)**.

Notably, the distilled model performs particularly well on **Nutrition (76.1% retention, 56.2% accuracy)** and **Professional Medicine (71.1% retention, 55.2% accuracy)**, which are among the most relevant to practical clinical applications. The strong performance on these tasks indicates successful transfer of clinical reasoning capabilities from the teacher model.

1) MedAlpaca Evaluation Results: Table II shows the performance on the MedAlpaca/medical-meadow-medical-flashcards dataset, which focuses on Direct Medical Question Answer.

TABLE II
PERFORMANCE ON MEDALPACA DATASET

Metric	**Llama3-Med42** 8B	**MedGemini** 4B	**LLaMA 3.2** 1B	**Distilled LLaMA1** Experiment 1	**Distilled LLaMA** Experiment 2
Rouge-1	0.3096	0.1777	0.1706	0.2119	0.2089
Rouge-2	0.1830	0.0934	0.0939	0.1393	0.1303
Rouge-L	0.2669	0.1493	0.1503	0.1793	0.1702
Rouge-Lsum	0.2675	0.1566	0.1547	0.1872	0.1772
BLEU Score	0.0711	0.0258	0.0264	0.0373	0.0343
BLEU-1	0.1551	0.0703	0.0777	0.0961	0.0941
BLEU-2	0.0818	0.0325	0.0350	0.0450	0.0430
BLEU-3	0.0532	0.0181	0.0178	0.0284	0.0254
prediction perplexity	3.1781	2.1432	2.6521	2.5514	2.6053
prediction avg loss	1.1562	0.7623	0.9751	0.8913	0.9138
token accuracy	0.5373	0.4822	0.4951	0.5051	0.5103

The distilled model achieves an average performance retention of **63.1%** of the teacher model's (**Llama3-Med42-8B**) capabilities on the MedAlpaca dataset, with particularly strong retention in **Rouge-2 (76.1%)** and **Rouge-Lsum (70.0%)** metrics. The best-performing distilled variant (**Distilled LLaMA - 1**) shows consistent improvements over both the baseline **LLaMA 3.2-1B (+27.8% relative improvement)** and **MedGemini-4B (+28.3% relative improvement)** across all metrics.

C. Manual Medical Expert Evaluation

An independent medical expert evaluated the Large Language Model (LLM) against reference answers, with the following key findings:

- **Comprehensiveness:** Most answers ($N = 70$) were more detailed and context-rich than the references.
- **Accuracy:** Many responses were medically correct and consistent with the reference.
- **Critical Errors:** Some cases ($N = 19$) showed serious factual mistakes, including misdiagnoses and incorrect mechanisms.
- **Terminological Precision:** Certain answers used informal or imprecise terms.
- **Omission of Key Information:** A few predictions missed essential conditions mentioned in the reference.

Overall, the LLM provides comprehensive explanations but remains prone to critical errors, emphasizing the need for expert oversight.

D. Illustrative Examples

The following examples illustrate the expert's evaluation process, contrasting a case where the LLM added valuable context (Table III) and a case where it introduced a significant error

E. Computational Efficiency Analysis

1) Model Size and Memory Requirements: Table IV compares the computational requirements of different models in the evaluation.

TABLE IV
COMPUTATIONAL EFFICIENCY COMPARISON

Model	Parameters (Billions)	Memory (GB)	Inference Speed (tokens/sec)	Latency (ms)
Llama3-Med42-8B	8.03	5.32	49.7	1006.04
MedGemini-4B	4.97	9.75	33.6	1486.60
LLaMA 3.2-1B (Base)	1.24	7.68	102.7	486.98
Distilled LLaMA 3.2-1B	1.24	7.68	59.5	841.02
Efficiency Gain	6.5×	0.69×	1.7×	0.58×

The distilled model maintains identical parameter count (1.24B) and memory requirements (7.68GB) compared to the base LLaMA 3.2-1B model, while achieving:

- **6.5× fewer parameters** than Llama3-Med42-8B (8.03B)
- **42.1% faster inference** than MedGemini-4B (59.5 vs 33.6 tokens/sec)
- **42.0% reduction in speed** compared to base LLaMA (59.5 vs 102.7 tokens/sec)

The latency increase compared to the base model (841.02ms vs 486.98ms) reflects the additional processing required for medical-specific reasoning patterns learned during distill.

F. Ablation Study

The results of the ablation study, presented in Table V, lead to several key insights:

1) **Effectiveness of Baseline KD:** The baseline knowledge distillation (Variant A) already provides a substantial improvement over the untrained LLaMA 3.2-1B model (39.6% on MMLU), achieving 42.3% accuracy. This confirms the fundamental capability of KD to transfer general medical knowledge.

TABLE III
EXAMPLES OF LLM PREDICTIONS WITH EXPERT EVALUATION.

Question	Reference Answer	LLM Prediction	Expert Opinion	Verdict
What are some possible causes of low PTH and high calcium levels?	PTH-independent hypercalcemia, which can be caused by cancer, granulomatous disease, or vitamin D intoxication.	Low PTH (parathyroid hormone) levels and high calcium levels may be seen in patients with primary hyperparathyroidism, caused by an overactive parathyroid gland. Other causes include high-dose vitamin D supplementation, vitamin D intoxication, sarcoidosis, and certain medications such as thiazide diuretics.	"The prediction elaborates on the causes of PTH-independent hypercalcemia and also includes other causes of low PTH and high calcium levels, making it a more comprehensive answer than the reference."	**Good**
What does low mobility and bulging of TM suggest?	Low mobility and bulging of TM is suggestive of Acute otitis media.	Low mobility and bulging of the tympanic membrane (TM) suggests a perforation of the TM, which may be due to a middle ear infection (otitis media) or other causes such as trauma, eustachian tube dysfunction, or chronic middle ear disease.	"There is a discrepancy. Reference states 'Acute otitis media,' while the prediction suggests 'perforation of the TM...'. The prediction lists potential causes, whereas the reference is a direct diagnosis."	**Bad**

2) **Impact of Progressive Temperature:** Introducing progressive temperature scheduling (Variant B) yielded a significant performance jump (+2.4% on MMLU). This demonstrates that gradually reducing the temperature softness is highly effective for medical distillation, as it first encourages the student to learn broad, nuanced relationships between conditions before sharpening its predictions for final accuracy.
3) **Contribution of Specialty Weighting:** The specialty-weighted loss (Variant C) provided further gains, particularly on the MMLU benchmark which is explicitly subdivided by specialty. This indicates that strategically balancing the focus across diverse medical domains is crucial for building a well-rounded clinical AI model.
4) **Importance of Attention Alignment:** Adding the attention alignment loss (Variant D, our full model) finalized our model's performance, contributing to an overall +5.4% gain over the baseline on MMLU. This underscores that transferring the *reasoning process*—i.e., *where* the teacher model looks for evidence—is just as important as transferring its final output distributions for complex clinical reasoning tasks. The attention loss ensures the student model learns to focus on clinically relevant cues in the input text, such as key symptoms or patient history details.

The ablation study conclusively validates that each component of our proposed framework contributes non-trivially to the final performance. The progressive strategy and attention alignment are particularly critical for distilling not just medical knowledge, but also the complex clinical reasoning patterns of the teacher model.

TABLE V
ABLATION STUDY OF THE PROPOSED DISTILLATION FRAMEWORK COMPONENTS. THE MMLU MEDICAL SCORE IS THE AVERAGE ACCURACY ACROSS ALL MEDICAL SUBTASKS.

Model Variant	MMLU Medical		MedAlpaca (Medical Meadow)			
	Acc. (%)	Δ	BLEU-1	Rouge-1	Rouge-2	Rouge-L
LLaMA 3.2-1B (Base)	39.6	–	0.218	0.281	0.161	0.213
A: Baseline KD	42.3	+2.7	0.225	0.292	0.168	0.221
B: A + Progressive Temp.	44.7	+5.1	0.234	0.305	0.176	0.235
C: B + Specialty Weight	46.5	+6.9	0.241	0.313	0.181	0.242
D: C + Attn. Align (**Distilled Model 1**)	**47.7**	**+8.1**	**0.247**	**0.321**	**0.185**	**0.247**

The ablation study (Table V) yields key insights:

1) **Baseline KD Effectiveness:** Confirmed KD's fundamental capability, with Variant A (42.3% MMLU) improving over the untrained base model (39.6%).
2) **Progressive Temperature Impact:** Variant B's scheduling caused a significant jump (+2.4% MMLU), showing its effectiveness for learning broad medical relationships before sharpening accuracy.
3) **Specialty Weighting Contribution:** Variant C provided further gains, indicating that balancing focus across medical domains is crucial for a well-rounded model.
4) **Attention Alignment Importance:** Our full model (Variant D) saw a +5.4% overall gain, underscoring that transferring the teacher's *reasoning process* (where it looks for evidence) is vital for complex clinical reasoning, ensuring the student focuses on clinically relevant cues.

The study validates that each component contributes non-trivially. Progressive strategy and attention alignment are critical for distilling both medical knowledge and complex clinical reasoning patterns.

VI. FUTURE WORK

The success of our approach opens several promising directions for future research:

- Investigation of multi-teacher distillation from complementary medical models to create more comprehensive student models.
- Development of dynamic distillation techniques that adapt to different medical specialties or clinical contexts.
- Integration of multimodal distillation (e.g., combining text with imaging, lab results, or EHR data) to enhance real-world diagnostic utility.
- Exploration of continual and lifelong distillation strategies that allow student models to incrementally acquire new medical knowledge without catastrophic forgetting.
- Design of hierarchical distillation pipelines, where knowledge is transferred in stages (e.g., from large generalist teachers to intermediate medical specialists, then to compact student models).
- Leveraging reinforcement learning–guided distillation, where the student is optimized not only for matching teacher outputs but also for clinical task-specific rewards.
- Development of safety-focused distillation techniques that explicitly incorporate medical safety constraints and

ethical considerations during the knowledge transfer process.

- Investigation of federated distillation approaches that enable collaborative model development across multiple healthcare institutions while preserving patient privacy.

VII. Conclusion

This paper presented a comprehensive knowledge distillation framework for creating efficient, high-performance medical language models. By distilling knowledge from both MedGemini-4B and Llama3-Med42-8B into LLaMA 3.2-1B, we demonstrated that it is possible to maintain 67.8% of the teacher model's accuracy on MMLU medical tasks while reducing the parameter count by 75%. The distilled model achieved particularly strong performance in clinically relevant areas such as Professional Medicine (71.1% retention) and Nutrition (76.1% retention), indicating successful transfer of practical medical knowledge.

In addition to accuracy-based benchmarks, the distilled model retained **81.0%** of the teacher's performance in prediction perplexity, **78.0%** in average loss, and an impressive **94.0%** in token accuracy. These results highlight the model's ability to not only preserve high-level reasoning skills but also maintain stable token-level generation quality.

Several key insights emerged from this work:

- Medical knowledge distillation benefits significantly from specialty-weighted loss functions and progressive temperature scheduling, which help preserve nuanced clinical reasoning patterns.
- Attention alignment proved crucial for maintaining the teacher model's focus on clinically relevant information during diagnosis and treatment recommendation tasks.
- The distilled model's 59.5 tokens/second inference speed makes it practical for real-time clinical applications, despite a 42% speed reduction compared to the base LLaMA model.

This work represents a significant step toward making advanced medical AI accessible in resource-constrained settings. By balancing model size, inference speed, and clinical accuracy, our distilled model demonstrates the feasibility of deploying high-quality medical language models in real-world healthcare environments. Future work should focus on clinical validation of these models in practical settings and development of specialized distillation techniques for particular medical use cases.

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References

[1] D. Wang and S. Zhang, "Large language models in medical and healthcare fields: applications, advances, and challenges," *Artificial intelligence review*, vol. 57, no. 11, p. 299, 2024.

[2] G. Hinton, O. Vinyals, and J. Dean, "Distilling the knowledge in a neural network," *arXiv preprint arXiv:1503.02531*, 2015.

[3] S. Hahn and H. Choi, "Self-knowledge distillation in natural language processing," *arXiv preprint arXiv:1908.01851*, 2019.

[4] X. Xu, M. Li, C. Tao, T. Shen, R. Cheng, J. Li, C. Xu, D. Tao, and T. Zhou, "A survey on knowledge distillation of large language models," *arXiv preprint arXiv:2402.13116*, 2024.

[5] T. Han, L. C. Adams, J.-M. Papaioannou, P. Grundmann, T. Oberhauser, A. Löser, D. Truhn, and K. K. Bressem, "Medalpaca–an open-source collection of medical conversational ai models and training data," *arXiv preprint arXiv:2304.08247*, 2023.

[6] J. Lee, W. Yoon, S. Kim, D. Kim, S. Kim, C. H. So, and J. Kang, "Biobert: a pre-trained biomedical language representation model for biomedical text mining," *Bioinformatics*, vol. 36, no. 4, pp. 1234–1240, 2020.

[7] K. Singhal, S. Azizi, T. Tu, S. S. Mahdavi, J. Wei, H. W. Chung, N. Scales, A. Tanwani, H. Cole-Lewis, S. Pfohl *et al.*, "Large language models encode clinical knowledge," *Nature*, vol. 620, no. 7972, pp. 172–180, 2023.

[8] C. Christophe, P. K. Kanithi, P. Munjal, T. Raha, N. Hayat, R. Rajan, A. Al-Mahrooqi, A. Gupta, M. U. Salman, G. Gosal *et al.*, "Med42–evaluating fine-tuning strategies for medical llms: full-parameter vs. parameter-efficient approaches," *arXiv preprint arXiv:2404.14779*, 2024.

[9] A. Sellergren, S. Kazemzadeh, T. Jaroensri, A. Kiraly, M. Traverse, T. Kohlberger, S. Xu, F. Jamil, C. Hughes, C. Lau *et al.*, "Medgemma technical report," *arXiv preprint arXiv:2507.05201*, 2025.

[10] S. Sreeram and B. Kainz, "Sculpting efficiency: Pruning medical imaging models for on-device inference," *arXiv preprint arXiv:2309.05090*, 2023.

[11] T. Dettmers, M. Lewis, Y. Belkada, and L. Zettlemoyer, "Gpt3. int8 (): 8-bit matrix multiplication for transformers at scale," *Advances in neural information processing systems*, vol. 35, pp. 30 318–30 332, 2022.

[12] Z. Su, H. Song, Y. Lin, Y. Wu, X. Weng, Z. Mao, B. Liu, H. Yin, and J. Yang, "Medkit: Multi-level feature distillation with knowledge injection for radiology report generation," *Expert Systems with Applications*, p. 129003, 2025.

[13] J. Wang, H. Ling, L. Zhang, L. Zhang, F. Wang, Y. Gao, and Z. Li, "Ckd-ehr: Clinical knowledge distillation for electronic health records," *arXiv preprint arXiv:2506.15118*, 2025.

[14] D. Hendrycks, C. Burns, S. Basart, A. Zou, M. Mazeika, D. Song, and J. Steinhardt, "Measuring massive multitask language understanding," *arXiv preprint arXiv:2009.03300*, 2020.

[15] D. Jin, E. Pan, N. Oufattole, W.-H. Weng, H. Fang, and P. Szolovits, "What disease does this patient have? a large-scale open domain question answering dataset from medical exams," *Applied Sciences*, vol. 11, no. 14, p. 6421, 2021.

[16] A. Pal, L. K. Umapathi, and M. Sankarasubbu, "Medmcqa: A large-scale multi-subject multi-choice dataset for medical domain question answering," in *Conference on health, inference, and learning*. PMLR, 2022, pp. 248–260.

[17] Q. Jin, B. Dhingra, Z. Liu, W. W. Cohen, and X. Lu, "Pubmedqa: A dataset for biomedical research question answering," *arXiv preprint arXiv:1909.06146*, 2019.

[18] T. Möller, A. Reina, R. Jayakumar, and M. Pietsch, "COVID-QA: A question answering dataset for COVID-19," in *Proceedings of the 1st Workshop on NLP for COVID-19 at ACL 2020*, K. Verspoor, K. B. Cohen, M. Dredze, E. Ferrara, J. May, R. Munro, C. Paris, and B. Wallace, Eds. Online: Association for Computational Linguistics, Jul. 2020. [Online]. Available: https://aclanthology.org/2020.nlpcovid19-acl.18/

[19] H. Yu, C. Yu, Z. Wang, D. Zou, and H. Qin, "Enhancing healthcare through large language models: A study on medical question answering," in *2024 IEEE 6th International Conference on Power, Intelligent Computing and Systems (ICPICS)*. IEEE, 2024, pp. 895–900.

[20] Y. Li, Z. Li, K. Zhang, R. Dan, S. Jiang, and Y. Zhang, "Chatdoctor: A medical chat model fine-tuned on a large language model meta-ai (llama) using medical domain knowledge," *Cureus*, vol. 15, no. 6, 2023.

[21] A. Johnson, T. J. Pollard, L. Shen, L.-w. H. Lehman, M. Feng, M. Ghassemi, B. Moody, P. Szolovits, L. A. Celi, and R. G. Mark, "Mimic-iii, a freely accessible critical care database, sci," *Data*, vol. 3, no. 1, pp. 1–9, 2016.

[22] A. Dubey, A. Jauhri, A. Pandey, A. Kadian, A. Al-Dahle, A. Letman, A. Mathur, A. Schelten, A. Yang, A. Fan *et al.*, "The llama 3 herd of models," *arXiv e-prints*, pp. arXiv–2407, 2024.